“Design and Implementation of a Secure Network Infrastructure for Al-Daraa Healthcare Group “

Case study:-

Al-Daraa Healthcare Group, one of Egypt’s leading healthcare providers, offers a wide range of medical services across its hospitals and clinics. With an expanding network of facilities in Cairo and Alexandria, the company has decided to implement its own network infrastructure to ensure enhanced control, security, and cost-effectiveness.

Previously, Al-Daraa relied on third-party service providers to maintain its IT infrastructure. However, considering the growing importance of digital healthcare solutions and the need for better data security, the senior management has decided to bring network management in-house. This decision necessitated a robust and secure network design capable of supporting all departments within the company while ensuring high availability, data integrity, and confidentiality.

The network design aims to connect two main locations: the headquarters in Cairo and a branch hospital in Alexandria, both of which are critical to the functioning of the company. The project will cover the implementation of Local Area Network (LAN) and Wide Area Network (WAN) structures, ensuring seamless communication between the various departments within both sites. The key components include the setup of VLANs for each department, configuration of secure inter-site communication through a VPN, and the use of Access Control Lists (ACLs) to manage access to different segments of the network.

This project will focus on creating a hierarchical network design with redundancy to ensure high availability, leveraging Cisco Packet Tracer to design and implement the network solution. Additionally, the solution will address key security measures such as port security, DHCP, DNS, web, and email server configurations, all of which will be hosted at the headquarters.

📍 **Headquarters Departments:**

1. **Medical Lead Operation & Consultancy Services (MLOCS)**
2. **Medical Emergency and Reporting (MER)**
3. **Medical Records Management (MRM)**
4. **Information Technology (IT)**
5. **Customer Service (CS)**
6. **Guest/Waiting Area (GWA)**

📍 **Branch Hospital Departments:**

1. **Nurses & Surgery Operations (NSO)**
2. **Hospital Labs (HL)**
3. **Human Resource (HR)**
4. **Marketing (MK)**
5. **Finance (FIN)**
6. **Guest/Waiting Area (GWA)**

Business Requirements:

1. **Cost-Effective Network**: The network must be affordable while maintaining high performance and scalability to meet future growth.
2. **Security**: The network must adhere to strict information security policies, including encryption of sensitive patient data and secure communication between the headquarters and the branch hospital.
3. **High Availability**: Redundancy should be built into the network to ensure that critical medical services are not interrupted due to network failures.
4. **Scalability**: The network should support future growth in terms of both user base and the addition of new departments or facilities.
5. **Ease of Management**: The network must be easy to manage and troubleshoot, with centralized control of critical network functions.

Functional and Non-Functional Requirements:

Functional Requirements:

1. **VLAN Configuration**: Each department in both HQ and the branch hospital should have a separate VLAN to ensure logical segmentation of traffic.
2. **Inter-VLAN Routing**: Communication between departments should be possible through multilayer switches configured for inter-VLAN routing.
3. **VPN**: A secure VPN tunnel must be implemented for communication between HQ and the branch.
4. **Dynamic IP Addressing**: DHCP servers must be configured to dynamically assign IP addresses to devices within the network.

5. **Access Control**: ACLs must be implemented to restrict access based on departments and roles.

Non-Functional Requirements:

1) **Reliability**: The network should be reliable and capable of maintaining uptime even in the case of hardware failures.
2) **Performance**: The network should be optimized for low latency and high throughput, especially for critical medical data.
3) **Security**: The network should comply with industry standards for healthcare data security, ensuring patient information is always protected.

LAN & WAN Structure Requirements:

1) **LAN Design**: The LAN will consist of multiple VLANs, each dedicated to specific departments such as IT, Medical Records, Customer Service, etc. The headquarters and branch hospital will each have one core router connected to two ISPs for redundancy. Multilayer switches will be used for inter-VLAN routing.

2) **WAN Design**: A secure, high-speed connection between the headquarters and branch hospital will be established using a VPN tunnel over the Internet. This will ensure encrypted communication between the two sites.
3) **Wireless Network**: A wireless network will be provided for each department to support mobile devices used by doctors, nurses, and other staff members.

Technical Requirements:

1) **IP Addressing**: The base network is 192.168.100.0. Subnetting will be applied to allocate appropriate IP ranges for each department.
2) **Routing Protocol**: BGP will be used as the routing protocol to allow dynamic route updates and maintain efficient routing between routers and multilayer switches across different networks.
3) **Firewall and ACLs**: ACLs will be configured to restrict access between departments based on their role. Additionally, a stateful firewall will be deployed for extra security.
4) **Security Protocols**: SSH will be used for secure management access to network devices. IPSec VPN will be used for encrypting communication between the HQ and the branch.
5) **Port Security**: Port security will be configured to limit access to the server-side switches and ensure that only authorized devices can connect to critical network segments.

Scope:

The project will cover the design and implementation of the network infrastructure for both the headquarters and the branch hospital. It will include:

- VLAN and subnetting design.
- Router and switch configuration, including inter-VLAN routing.
- VPN setup for secure communication between HQ and the branch.
- Configuration of DHCP, DNS, and other server-side services.

- Implementation of security measures such as ACLs, port security, and encryption.

Design & Implementation:

The network design will follow a hierarchical model, including:

- **Core Layer**: Routers connecting to the ISPs, providing redundancy for internet access.
- **Distribution Layer**: Multilayer switches for inter-VLAN routing.
- **Access Layer**: Access switches provide connectivity to devices within each department.

The implementation will include:

1. **IP Addressing and Subnetting**: Allocate IP address ranges to each department and configure routers and switches accordingly.
2. **VLAN Configuration**: Create VLANs for each department and assign switch ports to the appropriate VLANs.
3. **Routing Configuration:** Configure BGP and static routing on routers and multilayer switches.
4. **VPN Setup**: Implement a site-to-site IPSec VPN tunnel between HQ and the branch hospital.
5. **Security Configuration**: Set up ACLs, port security, and SSH for device management.
6. **Testing**: Verify the network's functionality, security, and performance.

Conclusion:

The successful implementation of this network infrastructure will significantly improve Al-Daraa Healthcare Group's operational efficiency, security, and ability to scale as the company grows. The use of VLANs, OSPF routing, VPN, and ACLs will ensure secure and efficient communication across the company's sites. With redundancy built into the network, the company can ensure high availability of critical medical services, which is essential in the healthcare industry. By investing in a robust network design, Al-Daraa Healthcare Group will be better positioned to meet the challenges of modern healthcare delivery while maintaining the highest standards of data security and patient care.